

Power Series and Taylor Series

ANSWER KEY



Radius and interval of convergence

- Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
Ratio test: $\left| \frac{x^{n+1}/(n+1)}{x^n/n} \right| \rightarrow |x|$. Converges for $|x| < 1$: radius of convergence $R = 1$.
- Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
Ratio test: $\left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \frac{|x|}{n+1} \rightarrow 0$ for every x : radius of convergence $R = \infty$.

Taylor and Maclaurin series

- Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n .
 $e^x = 1 + x + \frac{x^2}{2!} + \dots = 1 + x + \frac{x^2}{2} + \dots$
- Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.
 $f(0) = 1, f'(0) = 0, f''(0) = -1, f'''(0) = 0$: $T_3(x) = 1 - \frac{x^2}{2}$.
- Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.
 $0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001333\dots \approx 0.1987$.

Interval of convergence (including endpoints)

- Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.
Ratio test gives radius $R = 1$, centered at $x = 2$: $1 < x < 3$ tentatively. At $x = 3$: $\sum \frac{1}{n}$ diverges (harmonic).
At $x = 1$: $\sum \frac{(-1)^n}{n}$ converges (alternating). Interval of convergence: $[1, 3)$.
- Find the radius of convergence of $\sum_{n=1}^{\infty} n! x^n$.
Ratio test: $\left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| = (n+1)|x| \rightarrow \infty$ for any $x \neq 0$. The series converges only at $x = 0$: radius of convergence $R = 0$.

Using known series

- Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.
 $e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \dots = 1 - x^2 + \frac{x^4}{2} - \dots$
- Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).
 $\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \dots$

Taylor series centered at a nonzero point

- 10** Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.

$$T(x) = f(1) + f'(1)(x - 1) + \frac{f''(1)}{2!}(x - 1)^2 + \cdots = (x - 1) - \frac{(x - 1)^2}{2} + \cdots.$$

- 11** Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.

$$T_2(x) = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2.$$

Error estimation with Taylor polynomials

- 12** Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

$$T_2(0.3) = 1 - \frac{0.09}{2} = 1 - 0.045 = 0.9550, \text{ very close to the true value } 0.9553 \text{ (error } \approx 0.0003).$$